

Doc 10: Derivatives and Partial Derivatives

Module 2: Calculus | Part 1 of 3: Learn (theory). Take your time: every idea is built up slowly.

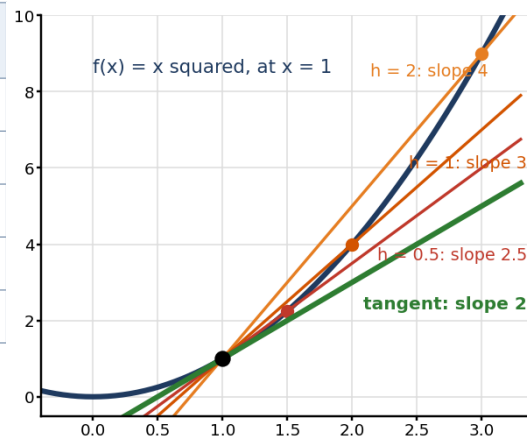
1. Slope of a curve: how fast is it changing right now?

In Doc 2 you found the slope of a **straight line**: rise over run, the same everywhere. A curve is steeper in some places and flatter in others, so its slope changes from point to point. The **derivative** is the slope of the curve **at one exact point**.

How to measure it: pick a small step h , draw a line through the curve at x and at $x + h$, and find its slope. Then shrink h . The slopes settle toward one number: the derivative.

h	Point $(1 + h, f)$	Slope of the line from $(1, 1)$
2	(3, 9)	$(9 - 1) / 2 = 4$
1	(2, 4)	$(4 - 1) / 1 = 3$
0.5	(1.5, 2.25)	$(2.25 - 1) / 0.5 = 2.5$
0.1	(1.1, 1.21)	$(1.21 - 1) / 0.1 = 2.1$
0.01	(1.01, 1.0201)	$0.0201 / 0.01 = 2.01$

For $f(x) = x^2$ at $x = 1$, the slopes 4, 3, 2.5, 2.1, 2.01 head toward 2. So the derivative of x^2 at $x = 1$ is 2. (Doc 10's rules will show it is $2x$, which is 2 at $x = 1$.)



$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Read it as: *the slope over a tiny step, as the step shrinks to zero*. **Notation:** $f'(x)$, df/dx and dy/dx all mean the derivative. You will rarely use the limit itself; the rules on the next page do the work.

2. What the derivative tells you

$f'(x) > 0$	the curve is rising at x	uphill
$f'(x) < 0$	the curve is falling at x	downhill
$f'(x) = 0$	flat at x : a possible valley or hilltop	candidate minimum or maximum
Large $ f'(x) $	steep	a small change in x changes f a lot

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Part 1 of 3: Learn (theory), continued

3. The rules: this short list gives you almost every derivative you need

Rule	Formula	Example
Constant	$d/dx (c) = 0$	$d/dx (7) = 0$
Power	$d/dx (x^n) = n x^{n-1}$	$d/dx (x^5) = 5x^4$
Power with negatives	same rule: rewrite first	$1/x = x^{-1}$ gives $-x^{-2}$
Power with roots	same rule: rewrite first	$\text{sqrt}(x) = x^{1/2}$ gives $\frac{1}{2} x^{-1/2}$
Constant multiple	$(c f)' = c f'$	$(3x^2)' = 6x$
Sum and difference	$(f \pm g)' = f' \pm g'$	$(x^3 - 4x)' = 3x^2 - 4$
Exponential	$d/dx (e^x) = e^x$	e^x is its own derivative
Natural log	$d/dx (\ln x) = 1/x$	
Product	$(f g)' = f' g + f g'$	$(x e^x)' = e^x + x e^x$
Quotient	$(f/g)' = (f' g - f g')/g^2$	$(x/(x+1))' = 1/(x+1)^2$

Power rule in words: bring the power down as a multiplier, then subtract 1 from the power. **Chain rule** (Doc 11) handles nested functions such as $(2x - 6)^2$; until then, expand the bracket first.

4. Tangent lines, minima, and the second derivative

Tangent line at $x = a$: $y = f(a) + f'(a)(x - a)$. It touches the curve at $x = a$ with the same slope. **To find a valley or hilltop:** solve $f'(x) = 0$. Then use the **second derivative** f'' (the derivative of the derivative): $f'' > 0$ means a valley (minimum), $f'' < 0$ means a hilltop (maximum).

Example: $f = x^2 - 4x + 3$. $f' = 2x - 4 = 0$ gives $x = 2$. $f(2) = -1$. $f'' = 2 > 0$, so $(2, -1)$ is a minimum.

5. Partial derivatives: several inputs, one slope at a time

When f depends on several inputs, for example $f(x, y)$, the **partial derivative** $\partial f/\partial x$ is the slope when you move x and **hold y fixed**: differentiate with respect to x and treat every other variable as a constant. $\partial f/\partial y$ does the reverse.

Example: $f = x^2y + 3y$. $\partial f/\partial x = 2xy$ (here y is just a number multiplying x^2). $\partial f/\partial y = x^2 + 3$ (here x^2 is just a number multiplying y). The list $(\partial f/\partial x, \partial f/\partial y)$ is the **gradient** (Doc 11).

6. Where machine learning uses derivatives

- **The loss** $L(w)$ measures error. dL/dw says how the error changes if the weight w changes a little.
- **Gradient descent** (Doc 13): move w opposite to the slope: w becomes $w - \eta \times L'(w)$. Example 7 shows it by hand.
- **ReLU** has derivative 0 for $x < 0$ and 1 for $x > 0$, which is why it passes gradients so cheaply.
- **Backpropagation** (Doc 24) is the chain rule applied over and over to get every weight's partial derivative.
- **Gradient checking:** compare a formula against $(f(x + h) - f(x)) / h$ for a tiny h , as in Exercise H.

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Part 2 of 3: Worked examples. Every cell filled in. Read, then cover the answers and redo them.

Example 1: From secant slopes to the derivative, $f(x) = x^2$ at $x = 3$

h	$f(3 + h)$	$f(3 + h) - f(3)$ ($f(3) = 9$)	Slope = difference / h
1	16	7	7
0.5	12.25	3.25	6.5
0.1	9.61	0.61	6.1
0.01	9.0601	0.0601	6.01

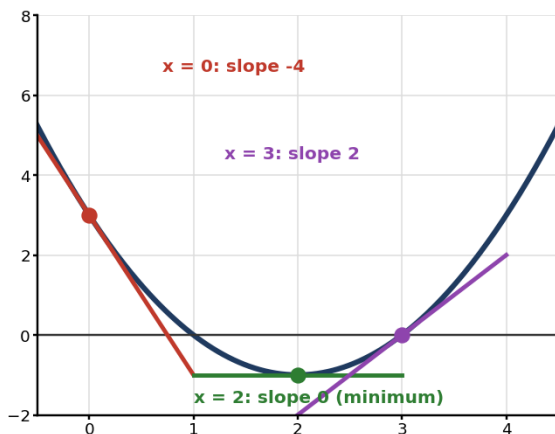
The slopes 7, 6.5, 6.1, 6.01 head toward **6**. The power rule agrees: $d/dx (x^2) = 2x$, and $2 \times 3 = 6$.

Example 2: Applying the rules, one term at a time

$f(x)$	Rewrite as powers (if needed)	$f'(x)$
$3x^4$	-	$12x^3$
$5x^2 - 2x + 7$	-	$10x - 2$
$1/x^2$	x^{-2}	$-2x^{-3} = -2/x^3$
$\text{sqrt}(x)$	$x^{1/2}$	$\frac{1}{2} x^{-1/2} = 1/(2 \text{ sqrt}(x))$
$(x + 1)^2$	$x^2 + 2x + 1$	$2x + 2$
$4e^x + 3 \ln x$	-	$4e^x + 3/x$

Row 2 term by term: $5x^2$ gives $5 \times 2x = 10x$; $-2x$ gives -2 ; the constant 7 gives 0. Row 3: $1/x^2 = x^{-2}$; bring down -2, then subtract 1 from the power: $-2x^{-3}$.

Example 3: Tangent lines and the minimum of $f(x) = x^2 - 4x + 3$



$$f'(x) = 2x - 4$$

x	$f(x)$	$f'(x)$
0	3	-4
1	0	-2
2	-1	0
3	0	2
4	3	4

Slope -4: falling steeply. Slope 0 at $x = 2$: the bottom of the valley. Slope 2: rising gently. $f'' = 2 > 0$, so $x = 2$ is a **minimum**, with value -1.

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Part 2 of 3: Worked examples, continued

Example 4: Product rule and quotient rule, step by step

Fill in u , v , u' , v' first. Then the formula does the rest. Product: $(u v)' = u' v + u v'$. Quotient: $(u / v)' = (u' v - u v') / v^2$.

Function	u	v	u'	v'	Result
$x^2 e^x$	x^2	e^x	$2x$	e^x	$2x e^x + x^2 e^x = x(2 + x) e^x$
$x / (x^2 + 1)$	x	$x^2 + 1$	1	$2x$	$(1(x^2+1) - x(2x)) / (x^2+1)^2 = (1 - x^2) / (x^2+1)^2$

Example 5: Partial derivatives read straight from a table of values

Take $f(x, y) = x^2 y$. The table lists its values. Move **along a row** (y fixed) to see the x -slope; move **down a column** (x fixed) to see the y -slope.

$y \backslash x$	$x = 0$	$x = 1$	$x = 2$
$y = 1$	0	1	4
$y = 2$	0	2	8
$y = 3$	0	3	12

At $(1, 2)$: row $y = 2$ reads 0, 2, 8. The slope from $x = 0$ to 1 is 2; from $x = 1$ to 2 it is 6. The average, 4, is $\partial f / \partial x$ at $x = 1$.

Rule check: $\partial f / \partial x = 2xy = 2 \times 1 \times 2 = 4 \checkmark$

Column $x = 1$ reads 1, 2, 3 for $y = 1, 2, 3$: the slope is 1 each step. Rule: $\partial f / \partial y = x^2 = 1 \checkmark$

Example 6: Partial derivatives by the rules, and a numeric check

$f(x, y) = x^2 y + 3y^2$.

Step	Work
$\partial f / \partial x$: treat y as a constant	$x^2 y$ gives $2xy$; $3y^2$ is a constant, so it gives 0. $\partial f / \partial x = 2xy$
$\partial f / \partial y$: treat x as a constant	$x^2 y$ gives x^2 ; $3y^2$ gives $6y$. $\partial f / \partial y = x^2 + 6y$
Evaluate at $(2, 1)$	$\partial f / \partial x = 2 \cdot 2 \cdot 1 = 4$. $\partial f / \partial y = 4 + 6 = 10$
Numeric check of $\partial f / \partial x$, $h = 0.01$	$f(2, 1) = 4 + 3 = 7$. $f(2.01, 1) = 4.0401 + 3 = 7.0401$. $(7.0401 - 7) / 0.01 = 4.01 \approx 4 \checkmark$

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Part 2 of 3: Worked examples, continued

Example 7: A first taste of gradient descent (Doc 13)

One data point says the best weight is $w = 4$, so the loss is $L(w) = (w - 4)^2 = w^2 - 8w + 16$. Its derivative is $L'(w) = 2w - 8$, which is 0 at $w = 4$. Start at $w = 0$ and repeat: **new $w = w - \eta \times L'(w)$** , with step size $\eta = 0.25$.

Step	w	Slope $L'(w) = 2w - 8$	New $w = w - 0.25 \times \text{slope}$
0	0	-8	$0 - 0.25(-8) = 2$
1	2	-4	$2 + 1 = 3$
2	3	-2	$3 + 0.5 = 3.5$
3	3.5	-1	$3.5 + 0.25 = 3.75$

The slope is negative (the loss falls as w rises), so w moves up. As w nears 4 the slope shrinks toward 0 and the steps shrink too: w closes in on the minimum at 4.

Common mistakes (each one is worth a second look)

Mistake	Correct
$d/dx (x^3) = x^2$ (forgot to bring the power down)	$3x^2$
$d/dx (x^3) = 3x^3$ (forgot to subtract 1 from the power)	$3x^2$
$d/dx (5) = 5$, or 1 (derivative of a constant)	0
$(f g)' = f' g'$ (product rule misremembered)	$(f g)' = f' g + f g'$
$d/dx (e^x) = x e^{x-1}$ (mixing up with the power rule)	e^x
$\partial f / \partial x$ of $x^2 y = 2x$ (dropped the y that acts as a constant multiplier)	$2xy$

Quick reference card

Power	$d/dx x^n = n x^{n-1}$	rewrite roots and fractions as powers first
Constant, multiple, sum	0; $(c f)' = c f'$; $(f \pm g)' = f' \pm g'$	term by term
e^x and $\ln x$	e^x ; $1/x$	
Product and quotient	$f' g + f g'$; $(f' g - f g') / g^2$	write u, v, u', v' first
Minimum or maximum	solve $f' = 0$; then $f'' > 0$ minimum, $f'' < 0$ maximum	
Tangent line	$y = f(a) + f'(a)(x - a)$	
Partial derivative	differentiate in one variable, treat the others as constants	$\partial f / \partial x, \partial f / \partial y$
Gradient descent	w becomes $w - \eta L'(w)$	Doc 13

Doc 10: Derivatives and Partial Derivatives

Part 3 of 3: Practice. Yellow cells are yours. Name: _____ Date: _____

A. Secant slopes closing in on the derivative

A1. $f(x) = x^2$ at $x = 2$, so $f(2) = 4$. Fill each row, then guess the derivative.

h	$f(2 + h)$	$f(2 + h) - 4$	Slope = difference / h
1			
0.5			
0.1			
0.01			

The slopes head toward _____. The rule $2x$ gives $2 \times 2 =$ _____.

A2. $f(x) = x^3$ at $x = 1$, so $f(1) = 1$.

h	$f(1 + h)$	$f(1 + h) - 1$	Slope
0.1			
0.01			

The slopes head toward _____. The rule $3x^2$ gives _____.

B. Power-rule warm-up

Hint: rewrite roots and fractions as powers first. Then bring the power down and subtract 1.

$f(x)$	Rewrite as powers (if needed)	$f'(x)$
x^5		
$3x^2$		
$7x$		
10		
$x^3 - 4x + 1$		
$2x^4 + x^2 - 9$		
$1 / x$		
$\text{sqrt}(x)$		
$5 / x^2$		
$(x + 2)^2$		
$e^x + 3 \ln x$		
$4e^x - 2x^3$		

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Practice, continued

C. Reading the derivative: $f(x) = x^3 - 3x$, $f'(x) = 3x^2 - 3$

x	-2	-1	0	1	2
f(x)					
f'(x)					
Rising, falling or flat?					

Where does $f'(x) = 0$ in the table? $x = \underline{\hspace{2cm}}$ and $x = \underline{\hspace{2cm}}$. Those are the hilltop and the valley.

D. Tangent line: $f(x) = x^2 + 1$ at $a = 2$

Step	Your answer
f(2)	
f'(x)	
f'(2)	
Tangent line: $y = f(2) + f'(2)(x - 2)$	
Tangent's prediction for f(2.1)	
True value $f(2.1) = 2.1^2 + 1$	

E. Minimum or maximum?

Function	f'(x)	x where f' = 0	Value there	f''(x)	Min or max?
$f = x^2 - 6x + 5$					
$g = -x^2 + 4x$					

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Practice, continued

F. Product and quotient rules

Fill u , v , u' , v' first, then apply the formula. Product: $u'v + uv'$. Quotient: $(u'v - uv') / v^2$.

Function	u	v	u'	v'	Result
1. $x e^x$					
2. $x^2 \ln x$					
3. $x / (x + 1)$					

G. Partial derivatives

Treat the other variable as a constant. Then evaluate at the given point.

Function and point	$\partial/\partial x$	$\partial/\partial y$	$\partial/\partial x$ at the point	$\partial/\partial y$ at the point
1. $f = 3x^2y + y^3 + 2x - 5$, point (1, 2)				
2. $g = x^2 + xy + y^2$, point (1, -1)				
3. $h = e^x y + \ln y$, point (0, 2)				

H. Numeric check: does the formula match $(f(x + h) - f(x)) / h$?

Check	$f(x + h)$	Slope over h	Rule value
x^3 at $x = 2$, $h = 0.1$			
x^3 at $x = 2$, $h = 0.01$			
$g = x^2 + xy + y^2$, $\partial g/\partial x$ at (1, -1), $h = 0.01$			

Hints: x^3 at $x = 2$ has $f(2) = 8$, and the rule gives $3x^2 = 12$. For g , $g(1, -1) = 1$ and the rule gives $2x + y = 1$.

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Practice, continued

I. Machine learning: slope of the loss and gradient descent

One data point has input $x = 2$ and target $y = 6$. With weight w , the loss is $L(w) = (2w - 6)^2$. Expand it first: $L(w) = 4w^2 - 24w + 36$.

I1. Find $L'(w) = \underline{\hspace{2cm}}$ and fill the table of slopes.

w	0	1	3	4
$L'(w) = 8w - 24$				

$L'(w) = 0$ at $w = \underline{\hspace{2cm}}$. That is the best weight ($2 \times \underline{\hspace{2cm}} = 6$).

I2. Gradient descent with step size $\eta = 0.1$: new $w = w - 0.1 \times L'(w)$. Start at $w = 0$.

Step	w	Slope $L'(w)$	New $w = w - 0.1 \times \text{slope}$
0			
1			
2			

Is w getting closer to the best weight? $\underline{\hspace{2cm}}$. Is the slope shrinking? $\underline{\hspace{2cm}}$

J. True or False?

Statement	True / False
1. If $f'(x) > 0$, the function is rising at x .	
2. If $f'(x) = 0$, x must be a minimum.	
3. The derivative of a constant is 0.	
4. For $\partial f / \partial x$, the other variables are treated as constants.	
5. The derivative of x^2 is x .	
6. The derivative of e^x is e^x .	

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Practice, continued: more repetitions (this is the skill that makes calculus click)

K. Eight more derivatives

Same routine: rewrite as powers or expand brackets if needed, then differentiate term by term.

$f(x)$	Rewrite (if needed)	$f'(x)$
$6x^2 - 5x + 2$		
$x^4 / 4 - 2x^2$		
$6 \sqrt{x}$		
$2 / x^3$		
$x(x + 3)$		
$e^x - \ln x$		
$(2x + 1)^2$		
$1 / (3x)$		

L. Rates of change in real life

L1. A ball's height in metres after t seconds is $h(t) = 20t - 5t^2$. Its speed is the derivative $h'(t) = 20 - 10t$ (positive = going up).

t (seconds)	0	1	2	3
$h(t) = 20t - 5t^2$				
$h'(t) = 20 - 10t$				
Up, top or down?				

The ball is at its highest point when $h'(t) = 0$, at $t = \underline{\hspace{2cm}}$, and its height then is $\underline{\hspace{2cm}}$ m.

L2. A loss is $L(w) = w^2 - 4w + 5$, so $L'(w) = 2w - 4$. To lower the loss, should w go up or down? (Slope negative: increase w . Slope positive: decrease w .)

w	0	2	5
$L(w)$			
$L'(w) = 2w - 4$			
Raise or lower w ?			

M. Self-check: how do you feel about each skill?

I can...	Revisit if unsure	Confident?
explain the derivative as the slope at a point	Example 1, Exercise A	
use the power rule (roots, fractions)	Example 2; B and K	
write a tangent line	Example 3, Exercise D	
find a minimum or maximum with f' and f''	Example 3, Exercise E	
use the product and quotient rules	Example 4, Exercise F	
take partial derivatives	Examples 5-6, Exercise G	
check a derivative numerically	Example 6, Exercise H	
take a gradient-descent step	Example 7, Exercise I	

Redo the exercises behind every 'Not yet' after a day or two: spaced practice beats one long session.

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Answer key (page 1 of 3): check your work only after finishing

A. Secant slopes

h	f(2 + h)	f(2 + h) - 4	Slope
1	9	5	5
0.5	6.25	2.25	4.5
0.1	4.41	0.41	4.1
0.01	4.0401	0.0401	4.01

Slopes head toward 4, and $2 \times 2 = 4$.

h	f(1 + h)	f(1 + h) - 1	Slope
0.1	1.331	0.331	3.31
0.01	1.030301	0.030301	3.0301

Slopes head toward 3, and $3 \times 1^2 = 3$.

B. Power-rule warm-up

f(x)	Rewrite	f'(x)
x^5	-	$5x^4$
$3x^2$	-	$6x$
$7x$	-	7
10	-	0
$x^3 - 4x + 1$	-	$3x^2 - 4$
$2x^4 + x^2 - 9$	-	$8x^3 + 2x$
$1 / x$	x^{-1}	$-x^{-2} = -1 / x^2$
$\text{sqrt}(x)$	$x^{1/2}$	$\frac{1}{2} x^{-1/2} = 1 / (2 \text{sqrt}(x))$
$5 / x^2$	$5x^{-2}$	$-10x^{-3} = -10 / x^3$
$(x + 2)^2$	$x^2 + 4x + 4$	$2x + 4$
$e^x + 3 \ln x$	-	$e^x + 3 / x$
$4e^x - 2x^3$	-	$4e^x - 6x^2$

C. Reading the derivative

x	-2	-1	0	1	2
f(x)	-2	2	0	-2	2
f'(x)	9	0	-3	0	9
Rising / falling / flat	rising	flat	falling	flat	rising

$f' = 0$ at $x = -1$ (hilltop, value 2) and $x = 1$ (valley, value -2).

D. Tangent line

Step	Answer
f(2)	5
f'(x)	$2x$
f'(2)	4
Tangent line: $y = f(2) + f'(2)(x - 2)$	$y = 5 + 4(x - 2) = 4x - 3$
Tangent's prediction for f(2.1)	5.4
True value $f(2.1) = 2.1^2 + 1$	5.41

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Answer key (page 2 of 3)

E. Minimum or maximum?

Function	$f'(x)$	x where $f' = 0$	Value	f''	Min or max?
$f = x^2 - 6x + 5$	$2x - 6$	3	-4	2	minimum
$g = -x^2 + 4x$	$-2x + 4$	2	4	-2	maximum

F. Product and quotient rules

Function	u	v	u'	v'	Result
1. $x e^x$	x	e^x	1	e^x	$e^x + x e^x = (1 + x) e^x$
2. $x^2 \ln x$	x^2	$\ln x$	$2x$	$1/x$	$2x \ln x + x^2(1/x) = 2x \ln x + x$
3. $x / (x + 1)$	x	$x + 1$	1	1	$(1(x+1) - x \cdot 1) / (x+1)^2 = 1 / (x+1)^2$

G. Partial derivatives

Function and point	$\partial/\partial x$	$\partial/\partial y$	at the point (x)	at the point (y)
1. $f = 3x^2y + y^3 + 2x - 5$, point (1, 2)	$6xy + 2$	$3x^2 + 3y^2$	14	15
2. $g = x^2 + xy + y^2$, point (1, -1)	$2x + y$	$x + 2y$	1	-1
3. $h = e^x y + \ln y$, point (0, 2)	$e^x y$	$e^x + 1/y$	2	1.5

Example 1: $\partial f/\partial x = 6xy + 2 = 6 \cdot 1 \cdot 2 + 2 = 14$ and $\partial f/\partial y = 3x^2 + 3y^2 = 3 + 12 = 15$. Example 3: $\partial h/\partial x = e^0 \cdot 2 = 2$ and $\partial h/\partial y = e^0 + 1/2 = 1.5$.

H. Numeric check

Check	$f(x + h)$	Slope over h	Rule value
x^3 at $x = 2$, $h = 0.1$	9.261	12.61	12
x^3 at $x = 2$, $h = 0.01$	8.120601	12.0601	12
$g = x^2 + xy + y^2$, $\partial g/\partial x$ at (1, -1), $h = 0.01$	1.0101	1.01	1

Each numeric slope is close to the rule value, and gets closer as h shrinks: 12.61, then 12.0601, heading to 12.

I. Loss slope and gradient descent

$L'(w) = 8w - 24$. Slopes at $w = 0, 1, 3, 4$: **-24, -16, 0, 8**. $L' = 0$ at $w = 3$ ($2 \times 3 = 6$).

Step	w	Slope $L'(w)$	New w
0	0	-24	2.4
1	2.4	-4.8	2.88
2	2.88	-0.96	2.976

w climbs $0 \rightarrow 2.4 \rightarrow 2.88 \rightarrow 2.976$, closing in on 3, and the slope shrinks in size (-24, -4.8, -0.96). $2.4: 8(2.4) - 24 = -4.8$; $2.88: 8(2.88) - 24 = -0.96$.

J. True or False?

1 True. 2 **False** (it could be a maximum or a flat spot). 3 True. 4 True. 5 **False** (it is $2x$). 6 True.

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Answer key (page 3 of 3)

K. Eight more derivatives

f(x)	Rewrite	f'(x)
$6x^2 - 5x + 2$	-	$12x - 5$
$x^4 / 4 - 2x^2$	-	$x^3 - 4x$
$6 \sqrt{x}$	$6x^{1/2}$	$3x^{-1/2} = 3 / \sqrt{x}$
$2 / x^3$	$2x^{-3}$	$-6x^{-4} = -6 / x^4$
$x(x + 3)$	$x^2 + 3x$	$2x + 3$
$e^x - \ln x$	-	$e^x - 1 / x$
$(2x + 1)^2$	$4x^2 + 4x + 1$	$8x + 4$
$1 / (3x)$	$(1/3) x^{-1}$	$-(1/3) x^{-2} = -1 / (3x^2)$

L. Rates of change

t (seconds)	0	1	2	3
h(t)	0	15	20	15
h'(t)	20	10	0	-10
Up / top / down	up	up	top	down

Highest point at t = 2 seconds (h' = 0), height 20 m. For h(1) = 20 - 5 = 15, h(2) = 40 - 20 = 20, h(3) = 60 - 45 = 15.

w	0	2	5
L(w)	5	1	10
L'(w)	-4	0	6
Raise or lower w?	raise	stay (minimum)	lower

L(0) = 5, L(2) = 4 - 8 + 5 = 1, L(5) = 25 - 20 + 5 = 10. The loss is smallest at w = 2, where the slope is 0.

M. Self-check

No wrong answers: any 'Not yet' points you to the example and exercise to redo. Aim to turn every row into a 'Yes' over the next few days.

Key takeaways: the derivative is the slope at a point (the limit of secant slopes); power rule: bring the power down, subtract 1; $f' = 0$ marks candidate minima and maxima, checked with f'' ; $\partial f / \partial x$ holds the other variables constant; gradient descent moves w by $-\eta \times \text{slope}$.

Next: Doc 11 - Gradients and the chain rule.